

Assignment-2:

1. Let G be a group of order pq , where p and q are distinct primes. Prove that G is abelian.

Ans: False.

For a group G of order pq to be abelian, a sufficient condition is that q does not divide $p-1$ (assuming $p < q$ without loss of generality) or p does not divide $q-1$.

Example: Dihedral group D_3 which has order

$$6 = 2 \cdot 3.$$

$$|D_3| = 6, \text{ so } p = 2 \text{ and } q = 3$$

D_3 is non-abelian. (rotation followed by

a reflection is not the same as a reflection followed by a rotation).

2 divides $3-1 = 2$, which is the condition under which a non-abelian group exists.

So, the statement is false.

2. If $|G| = p^r$ (p prime) then G is abelian if it has $p+1$ subgroups of order p .

Ans: If G is abelian, then G has $p+1$ subgroups of order p .

A group of order p^r must be abelian.

C_{p^2} (the cyclic group of order p^2).

C_{p^2} has exactly one subgroup of order p .

$C_p \times C_p$ (the direct product of two cyclic groups of order p).

The subgroups of order p in $C_p \times C_p$ are of the form $\langle (a, b) \rangle$ where (a, b) is not identity.

Total number of non identity elements is $p^2 - 1$.

Each subgroup of order p has $p-1$ non identity elements.

Since these subgroups intersect only at the identity, the number of subgroups is

$$\frac{p^2 - 1}{p - 1} = p + 1$$

Since G is a group of order p^2 , it must be abelian.

The center $Z(G)$ is non trivial ($|Z(G)| \in \{p, p^2\}$)

If $|Z(G)| = p^2$, G is abelian.

If $|Z(G)| = p$ then $G/Z(G)$ is cyclic of order

p . If $G/Z(G)$ is cyclic, G is abelian.

Thus, $Z(G) = p^2$ and G is abelian.

3. Let G be a finite group, and H be a proper subgroup of G . Prove: the union of all conjugates of H cannot be equal to G .

Ans: False

Let X be the set of left cosets of H in G ;

$|X| = [G:H] = n \geq 2$. G acts on X by left

multiplication. $\phi: G \rightarrow S_n$.

If $g \in G$ lies in some conjugate xHx^{-1} then

$x^{-1}gx \in H$, so g fixes the coset xH . Every $g \in G$ fixes at least one point of X . Hence

the image $\phi(G) \leq S_n$ is a subgroup in which

every element has a fixed point.

But a transitive subgroup of S_n always contains an element with no fixed points
a standard elementary lemma.

Since $\phi(G)$ is transitive, this contradicts

the every element has a fixed point

Conclusion: Therefore the union of conjugates of a proper subgroup H can't be all G .

4. Let G be a group and N be a normal subgroup of G . If G/N is cyclic and N is cyclic, prove G is abelian.

Ans: False.

Let's, Dihedral group $D_{2n} = \langle \pi, s \mid \pi^n = 1, s^2 = 1, s\pi s = \pi^{-1} \rangle$ with $n \geq 3$. The rotation subgroup $\langle \pi \rangle$ is cyclic and normal, and $D_{2n}/\langle \pi \rangle \cong C_2$ is cyclic, yet D_{2n} is nonabelian for $n \geq 3$.

so, the hypotheses do not force G to be abelian

5. Prove that in any group G , the set of elements of finite order form a subgroup of G .

Ans: False

Let, infinite dihedral $D_\infty = \langle \pi, s \mid s^2 = 1, s\pi s = \pi^{-1} \rangle$

Reflections $s\pi^k$ have order 2; the product of two distinct reflections $(s\pi^a)(s\pi^b) = \pi^{b-a}$ can have infinite order. Thus the product of two finite order elements need not have finite order. So, elements do not form a subgroup.

6. Let G be finite and p the smallest prime dividing $|G|$. Prove any subgroup of index p is normal.

Ans: True

Let $H \leq G$ with $[G:H] = p$. Let G act on the left cosets G/H by left multiplication.

$\varphi : G \rightarrow S_p$. The image $K = \varphi(G)$ is
 a subgroup of S_p . So, $|K|$ divides $p!$
 and also $|K|$ divides $|G|$. All prime
 divisors of $|K|$ are $\leq p$. By assumption
 p is smallest prime dividing $|G|$. So,
 every prime divisor of $|K|$ equals p . Hence
 $|K| = p^n$ is a power of p . But $K \leq S_p$
 and a p -subgroup of S_p has order at
 most p . So $|K| = 1$ or p .
 If $|K| = 1$ then φ is trivial and every
 g fixes every coset, so H is normal.